# Yassine Ben Maktouf

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**Junior Data Science Engineer and Actuary**

## Experience

- **Actuarial Data Scientist (End-of-Studies Internship) | Vermeg, Tunis** *Feb 2026 - Jul 2026*
  - Built a GenAI copilot that configures PERI retirement-savings products in Solife from specification documents, using an "extract then apply" design: an LLM extracts intent, then a deterministic, auditable engine (lxml/XPath, layered safety guards) applies each change to regulated XML and Groovy business rules.
  - Built a hybrid RAG assistant (all-MiniLM dense retrieval + BM25 with Reciprocal Rank Fusion) over an insurance corpus; benchmarked embedding models and retrieval strategies on MRR, Recall and NDCG to select the production configuration.
  - Compared fine-tuned XLM-RoBERTa with LLMs for named-entity extraction, audited faithfulness with NLI, and applied grammar-constrained decoding for reliable extraction; shipped a multi-instance React + FastAPI platform with JWT authentication and per-instance retrieval isolation.
- **Data Science Intern | Les Laboratoires MédiS, Nabeul** *Jun 2025 - Aug 2025*
  - Developed a sales forecasting system, comparing machine learning and deep learning models and tuning the best performer.
  - Built an automated SQL Server data pipeline and a web application for interactive reporting and visualisation.
- **Software Engineering Intern | STIP, Tunis** *Jun 2024 - Jul 2024*
  - Built a SQL Server database, Node.js REST APIs and a React interface to manage employees, phone numbers and contracts, with role-based access control.

## Key Projects

- **AI Platform for Medical Insurance Documents:** Processed handwritten French/Arabic medical documents with YOLO classification, fine-tuned OCR, zero-shot layout detection and a RAG-based LLM; added signature-based fraud detection (Siamese network) and reimbursement estimation. Deployed as microservices with FastAPI, Docker and Kafka; presented at ESPRIT's 12th Bal des Projets (2025).
- **Customer Churn Prediction:** Random Forest and Decision Tree on SMOTE-balanced data, PCA and K-Means to surface churn drivers; deployed with joblib-persisted scalers behind a REST API. github.com/YassineBenMaktouf/Customer-Churn
- **Audio Emotion Recognition:** Combined a diffusion model with a ResNet-50 classifier for speech emotion recognition on mel-spectrograms, evaluated on EmoDB and RAVDESS.
- **Mortality Modelling (R):** Lee-Carter model of French smokers' and non-smokers' mortality (1950 cohort) with trend projection and segmented annuity pricing.
- **Hackathon Projects:** Imaginify AI, an AI story-generation web app (1st place, Special AI Models Hack, lablab.ai, 2024); FinAssist, an AI personal-finance assistant that analyses Excel data (2nd place, ChatHack, lablab.ai, 2023); NASA International Space Apps Challenge (2024).

## Education

- **Master's in Actuarial Science (double degree) | IRA**, Le Mans University, France *2024 - 2026*
  Courses: risk theory, stochastic models, Monte Carlo methods, mathematical models for finance, life and non-life actuarial science, non-life reserving (Chain Ladder, Mack, Bootstrap), pricing (GLM), asset-liability management (ALM).
- **Engineering Degree in Computer Science, Data Science Specialty | ESPRIT**, Tunisia *2023 - 2026*
  Courses: advanced machine learning, deep learning and advanced deep learning, time series, big data analytics, MLOps and DevOps, optimisation for machine learning, probability and statistics, AWS cloud fundamentals.
- **Preparatory Classes in Mathematics and Physics | IPEIT**, Tunisia *2021 - 2023*
  Courses: analysis, linear algebra, probability, physics.
- **Baccalaureate in Mathematics | Lycée Ibnou Rachik**, Tunisia *2021*

## Skills & Certifications

**Machine learning & AI:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, YOLO, NLP, LLM, RAG, embeddings, retrieval evaluation
**Engineering & MLOps:** FastAPI, Flask, React, Docker, Kafka, MLflow, GitHub Actions, CI/CD
**Programming & data:** Python, SQL, R, Java, JavaScript, HTML, CSS, C#; SQL Server, MongoDB, MySQL, SQLite, vector databases
**Domain:** insurance and actuarial modelling: mortality, annuity pricing, risk theory, stochastic models, Solife (life and pensions platform)
**Certifications:** NVIDIA: Building Transformer-Based NLP Applications, Evaluation and Light Customization of LLMs, Applications of AI for Anomaly Detection; IBM: Machine Learning with Python, Deep Learning & Neural Networks with Keras, Introduction to Artificial Intelligence; AWS Academy Cloud Foundations
**Activities:** General Secretary of the ATAST club at ESPRIT (2023 - 2024), club ranked 2nd at the annual BTech competition; Member of the IEEE RAS (Robotics and Automation Society) student chapter at ESPRIT (2023 - 2024)
**Languages:** Arabic (native), English (bilingual), French (professional)